

Summary

MSc Computer Science student specializing in AI and Machine Learning, with experience in FPGA design, embedded systems, and software development (C/C++, Python). Eager to apply academic knowledge to real-world problems and further develop technical expertise.

Education

Aug 2025 – Present **Master's Programme in Computer Science**, *Linköping University*, Linköping, Sweden

Aug 2022 – Jun 2025 **Bachelor of Science in Computer Engineering**, *Linköping University*, Linköping, Sweden

Experience & Projects

- Mar 2025 – May 2025 **FPGA-based AES-128 Encryption (Bachelor Thesis)**, *Linköping University*, Linköping, Sweden
- Designed and implemented AES-128 encryption on both a **soft-core processor** and a **dedicated FPGA** using **VHDL**.
 - Compared performance in terms of **memory usage, latency, and throughput**, highlighting trade-offs in architectural approaches.
 - Used **simulation and synthesis tools** to evaluate and document results.
- Sep 2024 – Nov 2024 **Warehouse Robot – Project Course TSEA29**, *Linköping University*, Linköping, Sweden
- Developed a **mobile robot** for warehouse automation using **C/C++**, **AVR**, and **Raspberry Pi**.
 - Implemented **SPI** and **UART** communication and a **PD controller** for stability.
 - Designed **pathfinding algorithms** for autonomous navigation.
- Aug 2024 – Sep 2024 **Digital Oscilloscope – Project Course TSIU03**, *Linköping University*, Linköping, Sweden
- Built a **VHDL-based oscilloscope** to display live audio signals on a **VGA screen**.
 - Integrated **ADC input** for signal capture and **keyboard input** for control.
 - Utilized **Quartus software** for design and debugging.
- Oct 2022 – Nov 2023 **LIU Formula Student – AMS Engineer (Battery Management System)**, *Linköping University*, Linköping, Sweden
- Developed and maintained the **Battery Management System (BMS)** for an electric Formula Student car using **Arduino** and **STM32**.
 - Monitored **temperature, voltage, and current** to ensure performance and safety.
 - Contributed to **embedded software design, CAN communication**, and system reliability improvements.

Technical Skills

Programming & Scripting	C/C++, Python, JavaScript, VHDL, Node.js, Flask, React, HTML, CSS
Embedded Systems	STM32, Arduino, Raspberry Pi, SPI, UART, CAN, PD/PI Control, FPGA
Developer Tools	Git, GitHub, VS Code, Linux, Quartus, Simulation & Debugging Tools
Soft Skills	Teamwork, Analytical Thinking, Problem Solving, Technical Documentation

Additional Information

Languages Fluent in English; Fluent in Swedish